

```
fibonacci.cpp > main()
1  #include <iostream>
2  #include <stdexcept>
3  #include <vector>
4
5  std::vector<unsigned long long> fibonacciSequence(int n) {
6      if (n < 0) {
7          throw std::invalid_argument("n must be non-negative");
8      }
9
10     std::vector<unsigned long long> sequence;
11     sequence.reserve(static_cast<std::size_t>(n));
12
13     for (int i = 0; i < n; ++i) {
14         if (i == 0) {
15             sequence.push_back(0);
16         } else if (i == 1) {
17             sequence.push_back(1);
18         } else {
19             sequence.push_back(sequence[i - 1] + sequence[i - 2]);
20         }
21     }
22
23     return sequence;
24 }
25
26 int main() {
27     int n;
28     std::cin >> n;
29
30     std::cout << "Fibonacci sequence up to n = " << n << std::endl;
31
32     try {
33         const std::vector<unsigned long long> sequence = fibonacciSequence(n);
34
35         for (std::size_t i = 0; i < sequence.size(); ++i) {
36             if (i > 0) {
37                 std::cout << ' ';
38             }
39             std::cout << sequence[i];
40         }
41         std::cout << '\n';
42     } catch (const std::invalid_argument& error) {
43         std::cerr << error.what() << '\n';
44         return 1;
45     }
46
47     return 0;
48 }
```